

# Response to Reviewers

CVIP 2026 – Camera-Ready Submission

**Paper ID:** 258

**Paper Title:** Magnitude and Phase Aware Spectral Convolutional Neural Networks:  
Explainable Frequency-Domain Classification

We sincerely thank all the reviewers for their constructive feedback and insightful suggestions. Each comment has been carefully addressed in the revised camera-ready manuscript. Below, we provide a point-by-point response to every comment from each reviewer. All referenced changes are reflected in the revised manuscript.

## Reviewer 1

### Reviewer Comment:

*The paper addresses an important challenge in convolutional neural network (CNN)-based image analysis by incorporating frequency-domain representations and explainability mechanisms. The proposed spectral learning framework demonstrates potential significance, particularly in medical imaging applications where interpretability and transparency are essential for clinical trust. The integration of FFT-based representations and activation-gradient fusion appears technically interesting. The main problem with this paper is that the literature review appears to have limited coverage of recent research publications. The absence of sufficient references to the latest studies makes it difficult to adequately compare the proposed work with current state-of-the-art approaches and assess its relative contribution to recent advancements in the field.*

### Author Response:

In the revised manuscript, the Related Work section (Section 2) has been substantially restructured and expanded with **15 new references** spanning the period 2021–2025. These additions cover three key research directions that were previously underrepresented:

- (i) **Spectral architectures as core components:** We now discuss FcaNet [ICCV 2021], which generalises channel attention via the DCT; GFNet [NeurIPS 2021], which replaces self-attention with learnable global filters over the 2D Fourier spectrum; Adaptive Fourier Neural Operators (AFNO) [ICLR 2022] and Adaptive Frequency Filters [ICCV 2023] for efficient global token mixing. We also include WaveCNet [TIP 2021] for wavelet-based downsampling and FreqFusion [TPAMI 2024] for frequency-aware feature fusion in dense prediction.
- (ii) **Frequency-domain applications in vision:** We have added FreqNet [AAAI 2024] for frequency-aware deepfake detection, the survey [Neurocomputing 2022] on deep learning in the frequency domain, and the spectral bias study [ICML 2019] that formalises the observation that neural networks learn low-frequency functions first.
- (iii) **Explainability and medical imaging:** We now reference LayerCAM [TIP 2021] for hierarchical class activation maps, the comprehensive survey on XAI in medical image analysis [Medical Image Analysis 2022], the frequential-spatial saliency maps approach [Applied Sciences 2025], deformable large kernel attention [WACV 2024], and vision transformers in medical imaging [Medical Image Analysis 2023].

Furthermore, the Related Work now explicitly identifies **three concrete research gaps** against the cited state-of-the-art methods: (Gap 1) phase discarded or mishandled in existing spectral architectures, (Gap 2) single-level spectral views without jointly combining input-level and feature-level spectral analysis, and (Gap 3) lack of spectral explainability in CAM-style attribution methods, and positions our contributions as directly addressing all three. We believe this substantially strengthened survey now provides readers with adequate context to assess the relative contribution of our work.

## Reviewer 2

### Reviewer Comment:

*1. The contributions are not highlighted clearly in the introduction. The motivation and novelty of the work is not clear.*

### Author Response:

In the revised manuscript, the Introduction (Section 1) has been significantly restructured to improve clarity of motivation and novelty:

- **Motivation:** We now explicitly state the two core challenges upfront: (i) spectral inputs carry discriminative magnitude and phase information that spatial CNNs under-utilise, but feeding raw Fourier spectra to a network degrades accuracy because phase is highly sensitive to perturbations; and (ii) CAM-style explanation methods implicitly assume spatial inputs, making spectral model predictions unexplainable in a form clinicians can verify.
- **Novelty:** The contributions are now presented as a clearly numbered list of four items: (1) a magnitude and phase-aware 9-channel spectral input representation with sine-cosine phase encoding that is continuous and stable for learning; (2) a dual-frequency feature fusion (DFFF) module applying a second-level FFT to the deepest activation maps with adaptive channel-attention reweighting; (3) a frequency-domain explainability mechanism that computes CAM attributions on fused spectral features and maps them back to pixel space through inverse FFT; and (4) extensive evaluation across five CNN backbones on two datasets.
- **Bridge:** A new paragraph directly connects the clinical need for transparency in medical imaging to the technical gaps our framework addresses, clearly articulating why a unified spectral learning and explainability approach is necessary.

These changes make the novelty self-contained and immediately apparent.

### Reviewer Comment:

*2. The literature review needs more recent papers.*

### Author Response:

We agree. As detailed in our response to Reviewer 1, the Related Work (Section 2) has been expanded with **15 recent references** (2021–2025) covering spectral architectures (FcaNet, GFNet, AFNO, Adaptive Frequency Filters), wavelet-based methods (WaveCNet), frequency-aware fusion (FreqFusion), domain-specific applications (FreqNet for deepfakes, vision transformers in medical imaging), and advanced explainability techniques (LayerCAM, frequential-spatial saliency maps). Three concrete research gaps are now explicitly identified and positioned against these state-of-the-art methods to justify the novelty of our approach.

**Reviewer Comment:**

3. In the results, Table 1 and Table 2 are not clear. The proposed method is not mentioned and compared clearly.

**Author Response:**

In the revised manuscript, Table 1 and Table 2 have been clarified as follows:

- Each column heading now explicitly reads **Proposed (CNN backbone)** like **Proposed (ResNet-50)**, **Proposed (MobileNetV2)** etc., making it unambiguous that every column reports the performance of our proposed spectral framework instantiated with a different CNN backbone.
- The table captions have been revised to clearly state: *Performance of the proposed spectral framework on the [dataset] test set. Each column corresponds to the proposed method instantiated with a different CNN backbone.*
- The accompanying analysis text (Section 4, Quantitative Results Analysis) now explicitly contrasts the proposed method against pure spatial-domain CNNs, highlighting that the consistently high specificity and precision across both datasets and all five architectures demonstrate that the dual-frequency input reduces model confusion in regions where spatial features are ambiguous, a common limitation of spatial-domain CNNs in fine-grained and medical image classification.

These revisions ensure the reader can clearly identify what is being evaluated and understand the comparative context.

**Reviewer Comment:**

4. Ablation study is needed.

**Author Response:**

We have added a comprehensive ablation study in Section 4 of the revised manuscript (Table 3). Starting from a plain spatial-domain baseline, we incrementally enable three components using the best-performing backbone per dataset (ResNet-50 for Fruits-360, DenseNet-121 for Brain Tumor MRI):

| Configuration                             | Fruits-360 (%) | Brain Tumor MRI (%) |
|-------------------------------------------|----------------|---------------------|
| Spatial baseline (RGB only)               | 87.98          | 85.08               |
| + FFT magnitude input (3 channels)        | 88.75          | 85.66               |
| + Sine-cosine phase encoding (9 channels) | 89.18          | 86.23               |
| + DFFF module with channel attention      | <b>90.37</b>   | <b>87.62</b>        |

Three key observations emerge: (i) the magnitude spectrum alone improves over the spatial baseline, confirming that spectral energy carries discriminative information not readily accessible from raw pixels; (ii) sine-cosine phase encoding stabilises the perturbation-sensitive phase and adds localisation capability; (iii) the DFFF module yields the largest single improvement, demonstrating that re-examining deep activations in the frequency domain and adaptively reweighting the two streams is essential. The full configuration outperforms every reduced variant on both datasets.

**Reviewer Comment:**

5. Conclusion, future work needs improvement.

### Author Response:

The Conclusion and Future Work section (Section 5) has been substantially revised:

- **Conclusion:** We now provide a concise yet comprehensive summary that ties together the three main technical contributions: the stable sine-cosine phase encoding, the dual-frequency feature fusion module, and the frequency-domain explainability mechanism with the experimental evidence (up to 90.37% and 87.62% accuracy on Fruits-360 and Brain Tumor MRI, respectively), and the ablation results confirming that each component contributes complementary gains.
- **Future Work:** We articulate a concrete and actionable research agenda: (i) validation on larger clinical cohorts and additional modalities such as CT and histopathology; (ii) extension to transformer-based architectures; and (iii) development of a multi-resolution frequency pyramid ( $100\times 100$ ,  $50\times 50$ ,  $25\times 25$ ) with lightweight CNN processing and attention-based fusion, accompanied by a multi-scale CAM formulation for frequency-aware interpretability.

These improvements present a mature and forward-looking perspective on the work.

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### Reviewer 5

#### Reviewer Comment:

*1. The authors are encouraged to clearly highlight the motivation, novelty, and key contributions in the introduction of the paper. The literature review section also needs to be modified by positioning the identified research gaps against current state-of-the-art methods.*

### Author Response:

Both the Introduction and Related Work have been thoroughly revised:

**Introduction (Section 1):** The motivation now explicitly identifies two core challenges: (i) that spectral inputs carry discriminative magnitude and phase information but raw phase is too sensitive to perturbations for direct CNN consumption, and (ii) that CAM-style explanation methods assume spatial inputs, leaving spectral model predictions unexplainable. The four contributions are presented as a clearly numbered enumerated list immediately following the motivation paragraph, making the novelty self-evident.

**Related Work (Section 2):** The section has been expanded with 15 new references (2021–2025) and now concludes with an explicit positioning paragraph that identifies **three concrete research gaps** against the cited state of the art:

*Gap 1: **Phase discarded or mishandled** – Current spectral architectures (FcaNet, GFNet, Adaptive Frequency Filters) operate on magnitude-like quantities and do not exploit the localisation information in the Fourier phase.*

*Gap 2: **Single-level spectral views** – Existing methods apply frequency analysis either to the input or inside isolated layers, but do not jointly combine an explicit spectral input encoding with a second spectral analysis of the deepest features.*

*Gap 3: **Lack of spectral explainability** – CAM-style attribution methods assume spatial inputs and activations, so instance-level explanations for networks operating on Fourier-domain representations have received limited attention.*

Each gap is precisely mapped to the corresponding contribution of our framework (sine-cosine phase encoding → Gap 1; DFFF module → Gap 2; frequency-domain CAM with inverse FFT mapping → Gap 3), establishing a clear and direct justification for the work.

**Reviewer Comment:**

2. *The proposed method simply looks a hybrid combination of spatial-to-frequency and then dual-frequency feature fusion. The actual contribution needs to be highlighted with a strong mathematical reasoning. What authors propose as a main contribution is missing in the proposed method. The same also needs to be reflected in the experiments section. A comparison with more methods and providing reasoning for the performance gaps can help to better understand the proposed method and contributions.*

**Author Response:**

We deeply appreciate this critical observation. The concern that the method appears as a straightforward combination of existing components is valid on the surface, and we have taken significant steps to address it in the revised manuscript:

**(a) Mathematical reasoning for the phase encoding (Section 3.1):** We have added a formal mathematical justification for the sine-cosine phase encoding. Raw phase  $\Phi_c(u, v) \in (-\pi, \pi]$  lives on the circle  $S^1$ , so representing it as a scalar maps spectra with phases differing by an arbitrarily small amount near the branch cut to values separated by almost  $2\pi$ . A network consuming raw phase must therefore approximate a discontinuous function, making training unstable at the most informative frequencies. We prove that the mapping  $\Phi \mapsto (\cos \Phi, \sin \Phi)$  is an injective, 1-Lipschitz embedding of  $S^1$  into  $\mathbb{R}^2$  since:

$$\|(\cos \Phi_1, \sin \Phi_1) - (\cos \Phi_2, \sin \Phi_2)\|_2 = 2|\sin((\Phi_1 - \Phi_2)/2)| \leq |\Phi_1 - \Phi_2|$$

with components bounded in  $[-1, 1]$ . This guarantees small phase perturbations always produce proportionally small input changes, which is the precise property that makes phase learnable.

**(b) Mathematical reasoning for DFFF (Section 3.3):** The complementarity of the spatial and spectral streams is now proven formally. Convolutional activations  $A_k$  are translation-equivariant; by the Fourier shift theorem, their magnitude spectrum  $M_k^{(2)}$  is translation-invariant. By Parseval’s theorem, the spectral stream preserves total feature energy while reorganising it by spatial scale, and every spectral coefficient  $M_k^{(2)}(u, v)$  aggregates evidence from the entire feature map, giving each fused unit a global receptive field. The spatial stream encodes *where* structures occur while the spectral stream encodes *which* periodicities and scales are present, and neither can be recovered from the other once the magnitude operation discards phase. It is this mathematically grounded invariance–equivariance pairing with adaptive reweighting that constitutes the core contribution.

**(c) Ablation study demonstrating individual contributions (Section 4):** The new ablation study (Table 3) isolates the effect of each component by incrementally enabling them. The results quantitatively demonstrate that each component provides complementary and non-trivial gains:

- Spatial baseline (RGB only): The standard backbone trained on raw pixel inputs, serving as the reference point.
- FFT magnitude alone: Replacing RGB with 3-channel log-magnitude spectra, confirming spectral energy carries discriminative information not readily accessible from raw pixels.
- Sine-cosine phase encoding: Adding the cosine and sine phase channels on top of the magnitude further improves performance, validating the stable phase formulation.
- DFFF module: Enabling the dual-frequency feature fusion module produces the largest single improvement among all components, demonstrating that re-examining deep activations through a second-level FFT and adaptively reweighting the spatial and spectral streams via channel attention is essential for fully exploiting the spectral representation.

**(d) Comparison context in experiments:** The quantitative results discussion now explicitly contrasts the proposed approach against spatial-domain CNNs, explaining that the consistently high specificity and precision across both datasets and all five backbone architectures indicate that the dual-frequency input reduces model confusion in regions where spatial features are ambiguous. The qualitative analysis further demonstrates that spatial-domain CAM visualisations produce diffuse, poorly localised heatmaps, whereas the proposed framework produces structured activation maps that align with anatomical and pathological features.

We believe these additions: rigorous mathematical justification, formal proofs of complementarity, detailed ablation, and explicit performance reasoning, clearly distinguish our framework from a simple hybrid combination and demonstrate a principled, theoretically motivated contribution.

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We are grateful to all the reviewers for their thorough evaluation and valuable suggestions. Every comment has been carefully addressed in the revised camera-ready manuscript, and we are confident that the revisions substantially strengthen the paper. We hope the reviewers find the revised version satisfactory.

— *Naga Kushal Vankadara, Gokaramaiah Thota, Nagaraju K, Sathya Babu Korra, Viswanath Pulabaigari*